Thanh Khoi Nguyen

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Research Interests

I am broadly interested in **3D vision** and **multimodal understanding**, with a particular focus on **3D Gaussian Splatting** as a representation for scene understanding, object retrieval, and spatial reasoning. My work to date spans part-level 3D shape retrieval in GS representations, sports video analysis (athlete localization, field mapping, structured video reasoning), and vision-language models for multimodal content generation and retrieval. Going forward, I want to push toward **multimodal 3D scene understanding** - building systems that jointly reason over geometry, appearance, and language to enable richer analytics in dynamic, real-world environments.

Education

University of Science, VNU-HCM (HCMUS), B.Sc. in Computer Science — Sep 2023 – May 2027
Honor Programme - Information Technology Talent Program

- GPA: 3.94/4.0 - **Top 2%** of faculty; Dean's Honor List 2024 (Top 3%), 2025 (Top 2%)
- Admitted to the IT Talent Program (**top 45 students** selected faculty-wide by competitive examination)
- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Computer Systems, Computer Networks, Operating Systems, Database Systems, Mobile Engineering

Research Experience

Research Student | *Saigon AI Hub (SAIH), Applied AI Lab* — Early 2026 – Present
Co-founded by VNG Group & VNU-HCM · First globally selected for Google Labs' AI Futures Fund

- Selected into one of **10 inaugural research groups** at SAIH - Vietnam's first open AI research hub and the **first organization worldwide** chosen for Google Labs' AI Futures Fund, with access to early-release Google and DeepMind technologies.
- Conducting research on **multimodal sport reasoning and analysis**: building models that integrate visual, temporal, and language signals to understand and reason over complex sports scenarios in video.

Research Student | *Software Engineering Lab (SELab), HCMUS* — Jun 2024 – Present
Supervisor: Prof. Minh Triet Tran

- Conduct hands-on research in **3D vision and multimodal understanding**, working across 3D representation learning, video-based visual reasoning, and vision-language retrieval.
- Collaborate within a multi-project lab environment, contributing to researchs with publications at ECCV, ICMV, ICMR, and SOICT.
- Gained end-to-end research experience: problem formulation, experimental design, model implementation, and technical writing, under the supervision of Prof. Minh Triet Tran.

Publications

VISOR: View-Conditioned Spatial Reasoning for 3D Gaussian Referring Segmentation — Accepted, 2026
Thanh-Khoi Nguyen, Minh-Triet Tran*
Poster Track, SIGGRAPH Asia 2026 — *corresponding author

Action-Aligned Retrieval with Pairwise Multimodal Reranking for Text-Based Person Anomaly Search — Accepted, 2026
Thanh-Khoi Nguyen†, Thanh-Nhan Vo†, Trong-Thuan Nguyen, Minh-Triet Tran*
AI City Workshop @ ECCV 2026 — †equal contribution

A Top-Down Framework for Metric-Scale Athlete Localization from Single Broadcast Frames — Accepted, 2026
Thanh-Khoi Nguyen, Hoang-Phuc Nguyen, Linh-Huynh, Minh-Triet Tran*
Proc. ICMV 2026 (CORE C; SPIE-indexed)

TreeSoc: Tree-Structured Dynamic Reasoning and Tool Synergy for Soccer Video Understanding — Accepted, 2026
Thanh-Nhan Vo, **Thanh-Khoi Nguyen**, Trong-Thuan Nguyen, Trung-Hoang Le, Minh-Triet Tran*
Proc. ICMV 2026 (CORE C; SPIE-indexed)

SnapSeek 4.0: Bridging Temporal Dynamics and Regional Semantics for Interactive Lifelog Retrieval — Accepted, 2026

Minh-Quan Ho-Le, **Thanh-Khoi Nguyen**, Van-Tu Ninh, Cathal Gurrin, Minh-Triet Tran*
Proc. ACM LSC Workshop @ ICMR 2026 (CORE B)

From Text to Thumbnail: A Unified Framework for Automated News Image Generation and Evaluation Accepted, Nov 2025

Thanh-Khoi Nguyen, Hai-Dang Nguyen, Minh-Triet Tran*
Proc. SOICT 2025

Less is More: A Dual-Filtering Perspective on Prompt Design for News Thumbnail Generation Accepted, Oct 2025

Thanh-Khoi Nguyen, Hai-Dang Nguyen, Minh-Triet Tran*
CEUR Workshop Proceedings, MediaEval 2025

SHREC GS-3DORC: Towards the advancements of 3D Gaussian Splatting object part retrieval Published, Jun 2025

Thien-Phuc Tran, Minh-Quang Nguyen, **Thanh-Khoi Nguyen**, Nam-Quan Nguyen, Minh-Triet Tran*
Computers & Graphics (Elsevier, Q2), Vol. 132, 2025 10.1016/j.cag.2025.104321

Honors & Awards

- **1st Place, SoccerNet 2026 Spiideo SynLoc Challenge** 2026
  International sports AI benchmark for single-frame world-coordinate athlete localization; 97.67 mAP, surpassing baseline by over 21%.
- **Top 3, ACM MM 2026 TRIDENT Grand Challenge (Video Track)** 2026
  Tri-modal explainable deepfake forensics benchmark at ACM Multimedia (CORE A*); method GUARD achieved TCS 51.08 (detection 65.94, hallucination robustness 37.20, perception 45.13).
- **Best Paper Award, Youth Science Conference (YSC2026)** 2026
  Selected from all accepted submissions at the HCMUS university-level research conference.
- **Odon-Vallet Scholarship** 2023, 2026
  Nationally selective scholarship for exceptional academic merit in Vietnam.
- **Dean's Honor List** 2024 (Top 3%), 2025 (Top 2%)
  Highest-performing students in the faculty; two consecutive years.
- **Merit-Based Scholarship for Outstanding Students** 2023, 2024
  Awarded to top 5% of students faculty-wide; two consecutive years.
- **3rd Place, National Excellent Student Examination in Physics** 2023
  Top prize tier awarded to 12–15% of all national finalists across Vietnam.

Skills

Technical Skills: Scientific Research, Technical Writing, Problem Solving, Programming Logic, Machine Learning, Computer Vision
Programming Languages: C/C++, Python, Java, Kotlin, SQL, R, LaTeX, Markdown
Frameworks/Tools: PyTorch, TensorFlow, Keras, Hugging Face, Docker, Git, Android Studio, MySQL, PostgreSQL
Languages: Vietnamese (native speaker), English (professional working proficiency, IELTS 7.0)